



Intel Galileo board



Raspberry Pi board

Comparison between Galileo and Raspberry Pi Model B+

Parameters	Raspberry Pi Model B+	Intel Galileo
Price	US\$ 35	US\$ 70
Processor	700MHz ARM 11	400MHz Intel Quark
RAM	512MB	256MB
Storage	Micro SD card	Micro SD card
GPIO pins	26 GPIO pins	14 GPIO and six analogue input
Internet connectivity	Ethernet, compatible with USB Wi-Fi dongle	Ethernet, optional mPCIe card for Wi-Fi
Other important features	HDMI out, optional camera module	Arduino-compatible

IoT devices. These feature powerful ARM and x86 processors that allow connectivity to the Internet and also handle low-level hardware peripherals, sensors and actuators that engage with the environment. Let us see how these compare.

The hardware aspect

One of the main aspects of building connected IoT widgets is power consumption, since these are always on and often do not have access to mains power. Both Raspberry Pi and the Galileo have very low power consumption, consuming around 3W each. The rest of the power goes towards powering other sensors and peripherals. Low-level I2C, UART, GPIO and SPI bus peripherals are used by the main board to communicate with other hardware.

One of the main advantages of Intel Galileo is its Arduino compatibility. The Galileo has the same pin layout as Arduino, allowing it to use a large range of Arduino shields that have been developed since the first Arduino came out in 2005. It also gives users a very large, established code base, projects and a community.

A very large number of peripherals are made to be compatible with Arduino. All these factors make the Galileo a very attractive option both for hobbyists and firms wanting to get their product on the market quickly. This also gives a larger range of sensors and hardware for use in a widget.

Raspberry Pi, on the other hand, is a new entrant to the market, but has made a colossal splash in the market, getting extremely popular again with both the industrial and hobbyist crowd. It has developed a vast community, complete with compatible peripherals and code base. Though still lagging behind the Arduino family, it has emerged as a rival to the tried-and-tested Arduino. The Raspberry Pi Foundation has also encouraged the development of peripherals, such as Raspberry Pi camera module, which brings a 5MP HD camera at a price affordable for school children.

The software aspect

On the software side, the two boards are almost neck to neck. Microsoft has released a stripped-down light-

weight version of Windows for Intel Galileo. Microsoft's main goal with this push is twofold—first, to appease the growing hobbyist community as well as small and medium scale manufacturers, and second, to feed information collected by these IoT devices to Azure, its cloud service, and generate continuous revenues for the firm. Another plus-point of the Galileo is its Arduino compatibility, which means that it can run familiar Arduino codes, allowing users to port existing projects or build on thousands of open source projects found online, even though a Galileo-specific community is still in its nascent stage.

Raspberry Pi, on the other hand, leans more towards the open source side of things.

It can run Linux distributions such as Debian and Fedora, along with support for Android. Since it was originally designed for use in schools, it is designed to be extremely user-friendly and can run code written in Python and Scratch. Raspberry Pi also has a burgeoning dedicated user community online, which can be very helpful for debugging and code help.

Conclusion

Convergence of the IoT, Cloud, analytics and Big Data technology is the new trend in the embedded and software industry. These technologies together can be used to harness the tremendous power of data collected from thousands of Internet-connected sensors. Products like the IBM Internet of Things, Cloud and Windows Developer Program for IoT are evidence of the fact that software giants believe in the power of IoT and are scrambling for a share of the pie. The IoT is here to stay, and Raspberry Pi and Intel Galileo are products that can be leveraged to bring connected widgets to the market faster and cheaper than ever before. ●